Fiscal Year:	FY 2006	**Task Last Updated:**	FY 02/06/2007
PI Name:	Chatterjee, Aloke Ph.D.		
Project Title:	Simulated Space Radiation Studies for the Assessment of Chromosomal Damage: An Integrated Experimental and Theoretical Approach		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Berkeley	**State:**	CA
Zip Code:	94720	**Congressional District:**	9
Comments:	PI deceased as of June 2009.		
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-06
Start Date:	10/01/2002	**End Date:**	03/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: no-cost extension to 3/31/06, per PI 4-4-2006		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bedford, Joel (Colorado State University)		
Grant/Contract No.:	T-9466W		
Performance Goal No.:			
Performance Goal Text:			
	There are two main objectives of this research and they are:(1) experimental measurements of induced chromosomal aberrations by the mixed radiation field as a result of penetration of HZE particles through equivalent shielding materials as encountered in space travel and (20 development of a theoretical model which can determine how cell killing and chromosomal aberrations vary as a function of the thickness and composition of radiation shielding. In the process of designing spacecraft for long duration s pace flights, NASA may have to consider many combinations of thickness and composition of the shielding materials and it may be impractical to make measurements for every such combinatio, hence the existence of a reliable theoretical model will be of gre a t advantage in efficient spacecraft design. Chromosome aberrations are considered as surrogate end points for carcinogenesis. Such measurements in the blood lymphocytes of eight crew members were reported in the literature. This research directly contr i bu tes to at least two		

Task Description:	questions in the Critical Path Roadmap. One question relates to 10.08 under "Carcinigenesis caused by radiation" which states "how do the thickness,design and material composition of space vehicles affect the internal radiation en vi ron ment and biological assessment?" The second question relates to 10.09 which states that "what are the cancer risks in humans for spaceflight?" To produce a reliable algorithm based on experimental data which will allow to quantitatively assess the effects of different shielding materials and different thicknesses on HZE induced chromosome aberrations. It will help design spacecrafts to minimize the damaging effects of HZE particles.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research may be of great benefit in understanding the mechanisms associated with radiation induced carcinogenesis. In some institutions cancer patients are being treated with heavy ions and this research may ultimately provide information of great significance.
Task Progress:	It is clear from the Critical Path Roadmap that in order to understand risks associated with HZE induced carcinogenesis we must understand the radiation environment in which the astronauts will be operating inside the spacecraft. This radiation environment is dependent upon: (i) all the energetic components in the galactic cosmic rays and (ii) all the lower atomic number nuclear fragments that are generated as a result of nuclear disintegration as a single component of GCR penetrates through the shielding materials. The overall situation due to both of these sources of radiation is to present a very complex radiation environment. In order to estimate the carcinogenic potential of these radiation environments a systematic approach is required. This proposal relates to the effects of the fragments produced by the shielding. Once we understand the biological effects of the mixed beams acting simultaneously on cellular targets and consequent cellular and tissue responses, the approach could easily be extended to the situation where the incident radiation prior to the interaction with the shielding materials is a mixture of various radiation qualities as to be encountered in deep space. The theoretical model developed in this project can be used to predict biological effects such as chromosome aberrations for incident mixed beam combined with mixed beam due to nuclear disintegration caused by shielding materials. The effectiveness of the HZE iron ions for induction of acentric fragment producing chromosome-type aberrations at the first mitosis after irradiation of human GM2149 cells in G0 was not measurably different for incident Fe ion energies of 0.3, 0.6, or 1GeV/n. Neither did the filtration of 1 GeV/n or 0.6 GeV/n Fe ions with 1.8cm or 5.4 cm of aluminum or 5.5cm or 11.0cm of polyethylene measurably alter the effectiveness of the emerging beam. One of the most important conclusions derived from the successful completion of this project is that based on our current estimate of thicknesses of shielding materials, the biological effects of a mixed beam due to fragmentation processes will not be much different than the un-fragmented and pure beam. In addition, two manuscripts are in preparation for submission to Radiation Research.
Bibliography Type:	Description: (Last Updated: 06/24/2009)
Articles in Peer-reviewed Journals	Boissière A, Champion C, Touati A, Hervé du Penhoat MA, Sabatier L, Chatterjee A, Chetioui A. "DNA core ionization and cell inactivation." Radiat Res. 2007 Apr;167(4):493-500. PubMed PMID: 17388690 , Apr-2007
Articles in Peer-reviewed Journals	Asaithamby A, Uematsu N, Chatterjee A, Story MD, Burma S, Chen DJ. "Repair of HZE-particle-induced DNA double-strand breaks in normal human fibroblasts." Radiat Res. 2008 Apr;169(4):437-46. PubMed PMID: 18363429 , Apr-2008